

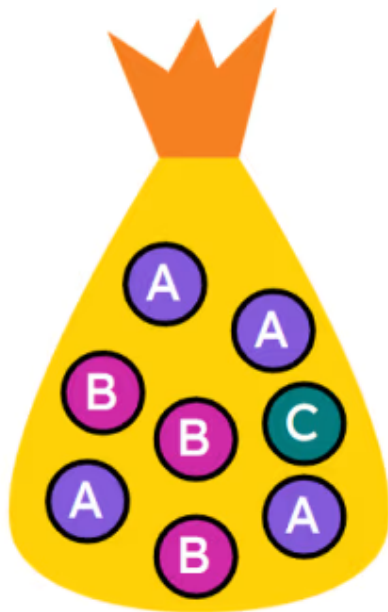


Calculating theoretical probabilities from two-way tables (two events)

- 1 In a fair six-sided die, what is the probability of rolling a 5 and getting a head in a single toss of a fair coin? Tick **1** correct answer

- ☐ $\frac{1}{2}$
- ☐ $\frac{1}{6}$
- ☒ $\frac{1}{12}$

- 2 A counter is taken from the bag. Its letter is noted and then it is placed back into the bag. A counter is taken from the bag again. Which of these options is not the correct sample space? Tick **1** correct answer



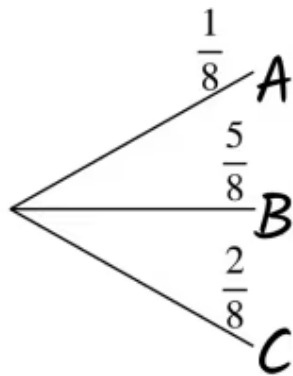
- ☐ $\xi = \{AA, AB, AC, BA, BB, BC, CA, CB, CC\}$
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- 3 If the probability of an event (A) is $\frac{3}{7}$, and the experiment is repeated 49 times, what is the theoretical number of times event (A) will occur? Tick **1** correct answer

- ☐ 3
- ☐ 7
- ☒ 21
- ☐ 49

- 4 Which of the events {A,B,C} is the least likely to occur according to this probability tree? Tick **1** correct answer



- ☒ A
- ☐ B
- ☐ C
- ☐ None.

- 5 John uses an unbiased random selection tool to generate a single letter of the alphabet. It is repeated three times. Which of the following letter sequences is the least likely? Tick **1** correct answer

- ☐ A,G,H
- ☐ A,A,A
- ☐ M,O,O
- ☐ B,I,N
- ☒ All selections are equally likely

- 6 There are two trials. In Trial 1, a spinner with {A, B} is spun twice. In Trial 2, a spinner with {B, C} is spun twice. Which statement is true? Tick **1** correct answer

- ☐ The total number of possible outcomes in Trial 1 is greater than in Trial 2.
- ☐ The total number of possible outcomes in Trial 1 is less than in Trial 2.
- ☒ The total number of possible outcomes in Trial 1 is the same as in Trial 2.



